Tan Dai Ngo

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Technical Skills

- **Programming:** Python, Java, SQL, MATLAB
- **Machine Learning:** scikit-learn, XGBoost, CatBoost, regression, classification, clustering, feature engineering, Optuna
- **Data & ML Systems:** Apache Spark, Kafka, Cassandra, Neo4j, MLflow, ETL pipelines, distributed systems
- **Deployment & GenAI:** FastAPI, Docker, AWS (EC2), Sentence Transformers, Qdrant, RAG, Ollama LLM

Education

Master of Engineering in Applied Data Science | GPA: 8.68 / 9.00 — *University of Victoria | Sep 2025 - Dec 2027 (Expected)*

- **Coursework:** Optimization for ML, Systems for Massive Datasets, Data Analysis & Pattern Recognition, Data Mining, Algorithms & Data Models, Artificial Intelligence

Master of Science in Computer Science — *University of Chicago | Sep 2020 - Mar 2022*

- **Coursework:** Algorithms, Distributed Systems, Advanced Python Programming, Databases, Applied Software Engineering

Bachelor of Science in Economics | GPA: 3.57 / 4.00 — *University of Washington | Sep 2015 - Aug 2019*

- **Minors:** Informatics, Applied Mathematics

Experience

Math & Computer Science Teacher | APU American International Schools — *Da Nang City, Vietnam | Aug 2024 - Dec 2024*

- Taught Python programming, algorithms, OOP, and software engineering fundamentals to 50+ students through hands-on coding projects with focus on debugging, algorithmic reasoning, and performance optimization.

Software Developer | T-Mobile (via BeaconFire Inc.) — *Bellevue, WA, USA | Jun 2022 - Apr 2024*

- Automated Kafka reprocessing pipelines for 50+ weekly roaming service tests, cutting manual intervention by 80%.
- Redesigned Cassandra schemas for partner/workflow microservices (100k+ records), maintaining query latency under 3s.
- Developed Spring Boot microservices integrated with Jenkins CI/CD and Splunk, enabling scalable downstream analytics.

Projects

Distributed Ocean Data ML Platform with RAG — *Personal Project*

- Built an end-to-end distributed ML pipeline using Apache Spark (Bronze → Silver → Gold architecture) to ingest and harmonize 10M+ NOAA and ONC ocean sensor observations for scalable feature engineering and time series forecasting.
- Developed and tuned XGBoost forecasting models using Optuna and MLflow, applying chronological holdout validation on temporally correlated ocean data (RMSE: 0.00586, R^2: 0.9999).
- Deployed FastAPI services for model inference and RAG search using Sentence Transformers, Qdrant, and Ollama with retrieval benchmarking (hit@k: 1.0, term recall: 0.875) and a Streamlit dashboard for interactive exploration.

End-to-End Machine Learning System for Multi-Output Fuel Blending — *Shell.ai Hackathon 2025*

- Built and deployed a production ML system for multi-output regression across 10 chemical blend properties using FastAPI, Docker, and AWS EC2 for real-time inference.
- Benchmarked XGBoost and CatBoost using 5-fold cross-validation, selecting CatBoost after achieving significantly lower error (MAPE: 0.64 vs. 1.29).
- Designed feature engineering and inference pipelines using weighted aggregation, mixture metrics based on entropy, and serialized preprocessing to ensure consistent production predictions.

Anomaly Detection with Isolation Forest on High Volume Bird Observation Data (Team of 3) — *Systems for Massive Datasets*

- Built an anomaly detection system on a distributed eBird dataset (1M+ observations) using a custom implementation of Isolation Forest with data processing in PySpark.
- Designed an end-to-end ML pipeline including feature engineering (species frequency, geospatial features, cyclical time encoding) and stratified sampling to preserve data distribution.
- Discovered a geographic variant of feature-specific swamping in Isolation Forest, where skewed spatial distributions caused the model to miss anomalies in dense regions, and proposed rank transformation and density-aware subsampling as fixes.